Progress Project Report

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Senior Design II

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The Portable Emissions Tracer

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- **List of Tables and Figures**
- **Introduction**
  - Answer to: (Roughly 2-3 sentences each)
    - What is the problem being solved?
    - Why is the problem important to solve?
    - Is the problem still unsolved? Have previous attempts been made?
    - Why is the problem difficult to solve?
    - How have we solved the problem?
    - What are the conditions under which our solution is applicable?
    - What are the main results?
    - What is the summary of our contributions?
    - How are the results of the report organized?
- **Detailed Background and Past Related Work**
- **Project Management**
  - Accurate Summary of Project aka Gantt Chart
  - Prototype Test Plan (Final Version)
- **Technical Section/Project Deliverables** (All for final version)
  - Design Detail
  - Calculations
  - Use of Figures
  - Terminology
  - Modeling
  - FEA
  - Control Sequence
  - Etc...
    - Note: All other data for other versions of protype go into Appendix
- **Project Results and Analysis**
- **Challenges Faced**
- **Final Budget**
  - Mass Production Calculation
- **Recommendations and Future Work**
- **Conclusion** (About 50-100 words)
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Abstract

Additive manufacturing processes release volatile organic compounds (VOCs), aerosols, and ultrafine particles that can accumulate in poorly ventilated environments and contribute to long-term respiratory or cardiovascular diseases. Industrial systems often include filtration or monitoring equipment, but affordable, real-time solutions are rarely available for non-industrial users such as hobbyists, students, and small labs.

We propose a **portable handheld emissions tracer** that integrates multiple sensors for ultrafine particulate matter (down to 1.0 µm), total VOCs, and formaldehyde into a compact, battery-powered platform. The device provides up to four hours of operation, can function as either a handheld or stationary monitor, and enables remote access to air-quality data through a web interface.

Our research involved literature review, analysis of OSHA standards, and consultation with a campus 3D-printing lab. Based on these findings, we developed a sensor architecture that delivers accurate, real-time monitoring of hazardous emissions. By providing accessible alerts and data visualization, the system empowers non-industrial users to make informed decisions, reduce exposure risks, and improve additive-manufacturing safety.

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Introduction

As 3D printing technology becomes more affordable and accessible, concerns about its impact on operator health have grown. Both Fused Deposition Modeling (FDM) and Stereolithography (SLA) resin printers emit volatile organic compounds (VOCs), aerosols, and particulate matter during printing and post-processing. Research conducted by Chemical Insight indicates that prolonged exposure to these emissions can contribute to respiratory issues, skin irritation, and other health concerns (Chemical Insight, p.2). However, because consumer 3D printing has only become widespread in the past decade, many personal and academic users remain unaware of these risks and often lack proper ventilation or emission monitoring.

Existing mitigation methods primarily rely on ventilating the area or using HEPA and carbon filters to capture hazardous particles and gases. While effective, there are limited ways for users to monitor their 3D printing environment, one of which is generic air quality monitoring on the market. Although general-purpose air quality monitors are commercially available, most are designed for ambient indoor environments rather than the specific emission profiles of 3D printing. They often lack the sensitivity, calibration, or sensor range needed to accurately detect VOCs, ultrafine particles (UFP), and specific gas such as formaldehyde produced by FDM and SLA printers. To address these limitations, our project proposes a portable emission tracer—a compact, sensor-based monitoring system tailored specifically for 3D printing environments to detect and alert users of hazardous emission levels. The device integrates multiple gas sensors connected to a Raspberry Pi Zero 2, using an IoT framework to collect and transmit emission data to a server for real-time monitoring. This system provides an affordable, user-friendly solution that promotes safe operation in academic labs, maker spaces, and home workshops while complying with OSHA and ANSI safety standards. The current scope of the air monitor can adequately collect data (concentration of VOC'S in the environment) and display the results on the screen. The necessary leftovers will be required to do the 3D printing of the housing to protect the electronic system and make it accessible to portability.

Background

3D Printing, whilst an immensely powerful tool, does not come without drawbacks. The use of this technology can cause the buildup of harmful substances that when inhaled can lead to long term health problems as stated by the CDC [2]. Multiple studies have confirmed that desktop 3D printing processes generate both volatile organic compounds (VOCs) and particulate matter, especially ultrafine particles (UFPs) [3], [4], [6], [10], [12]. In an article by Azimi published in 2016, UFP and VOCs emission is quantified, and it is suggested that users need to be caution when using 3D printers in poorly ventilated spaces, or without a proper filtration system [4].

This technology has been around for a short while, even shorter for commercial use. Many who are already in industry are aware of the risks that are involved when working with diverse types of machines and the different hazards associated with them. The bigger issue though is that many who use 3D printers commercially are ignorant of the possible ramifications of continuous use.

The main approach to helping keep atmospheric conditions acceptable is through ventilation and proper air flow within a system. This does not guarantee that a work or living environment is safe to breathe in even with an active ventilation system present. To help monitor this, sensors for common chemical offputs like formaldehyde were developed as well as sensors for particulate matter and ultrafine particles (UFP). Most of these sensors are used in settings such as industrial complexes and laboratories and are not easily accessible to ordinary households.

Table 1: Comparison to two existing Products

Device/ Aspect	Description	Limitations	Key Insight for Our Design
Existing Solutions	Various air quality monitors exist for home and industrial use	Often stationary, expensive, or non-IOT based	Need for portable, affordable IOT-integrated design
AirGradient ONE	Open-source PM monitor detecting VOCs, Wi-Fi and support	Fixed setup requires assembly, not handheld.	Inspires our open-source and IOT-based approach
Temtop	Commercial handheld device detecting gases and particulates	Costly, closed source, lacks IOT integration,	Motivates our portable, cost-effective, IOT-enabled design
Our Approach	Combines AirGradient's openness and Temtop's sensor variety	Using Raspberry Pi zero 2, limiting the duration of use	Adds IOT monitoring, rechargeable power

Tables and Figures

Since working with hazardous gases is required, there are exposure limits according to safety standards from organizations such as OSHA, CDC, and NIOHS. For particulate matter, as stated by the CDC in their Particulate Matter Guidance, assessment of PM exposure to establish an acceptable range is complex; they based it on WHO and the results for PM exposure limit are in Table 2 (CDC PM). In Table 1 the common VOCs that are produced during 3D printing processes, and Table 3 highlights the most toxic gases and hardest for humans to detect because they are odorless. The two graphs below show how much VOCs and particulate matter, size and concentration are emitted during the printing process. For figure 2, we can see that at the beginning of the printing process, most of the particles are between the size of 7 to 50 nm, but these smaller particles will combine to form larger particles leading to a significant drop of the amount of the smaller size particles.

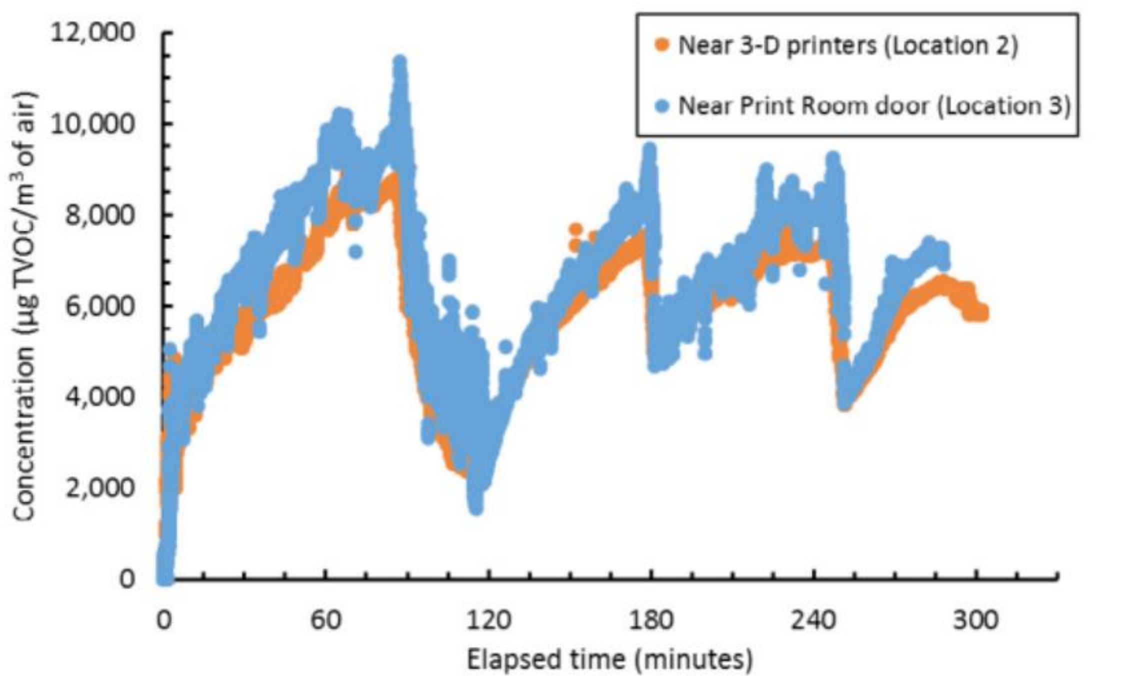


Figure 1: **Reprinted from Volatile Organic Compounds during printing processes** NIOSH [2017]

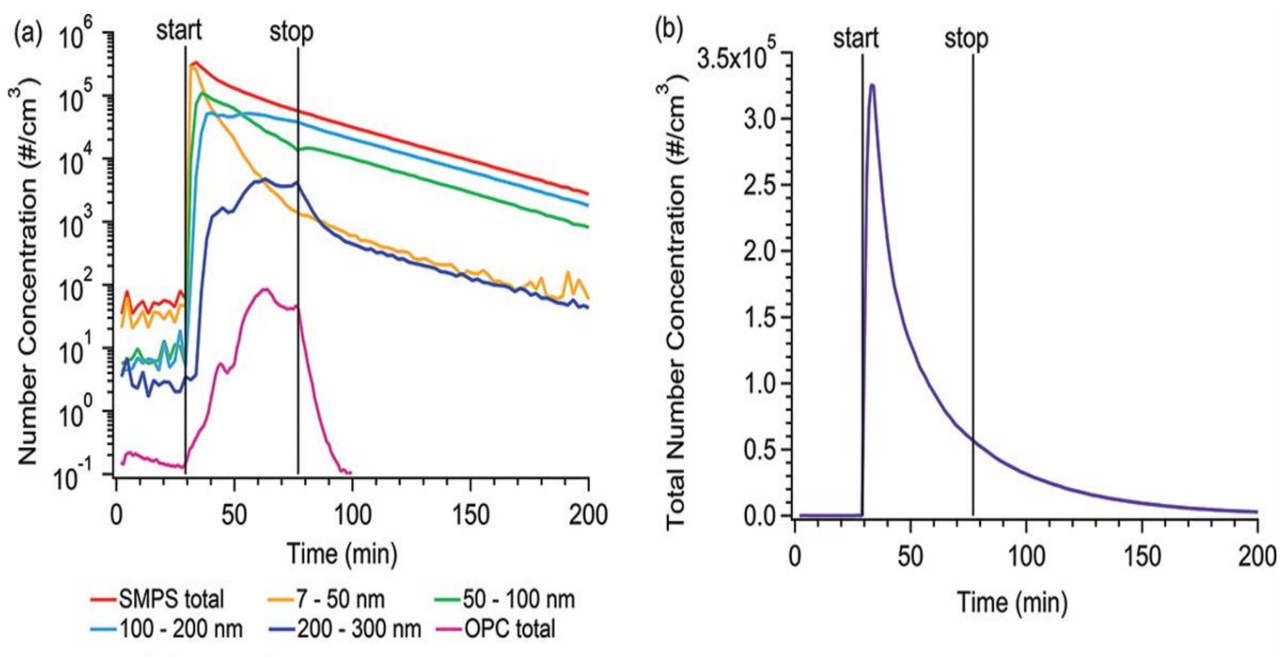


Figure 2: **Particle size distribution in the Print Room during the day while printing only with PLA filaments.** The average particle size was about 44 nm**. [HHE]**

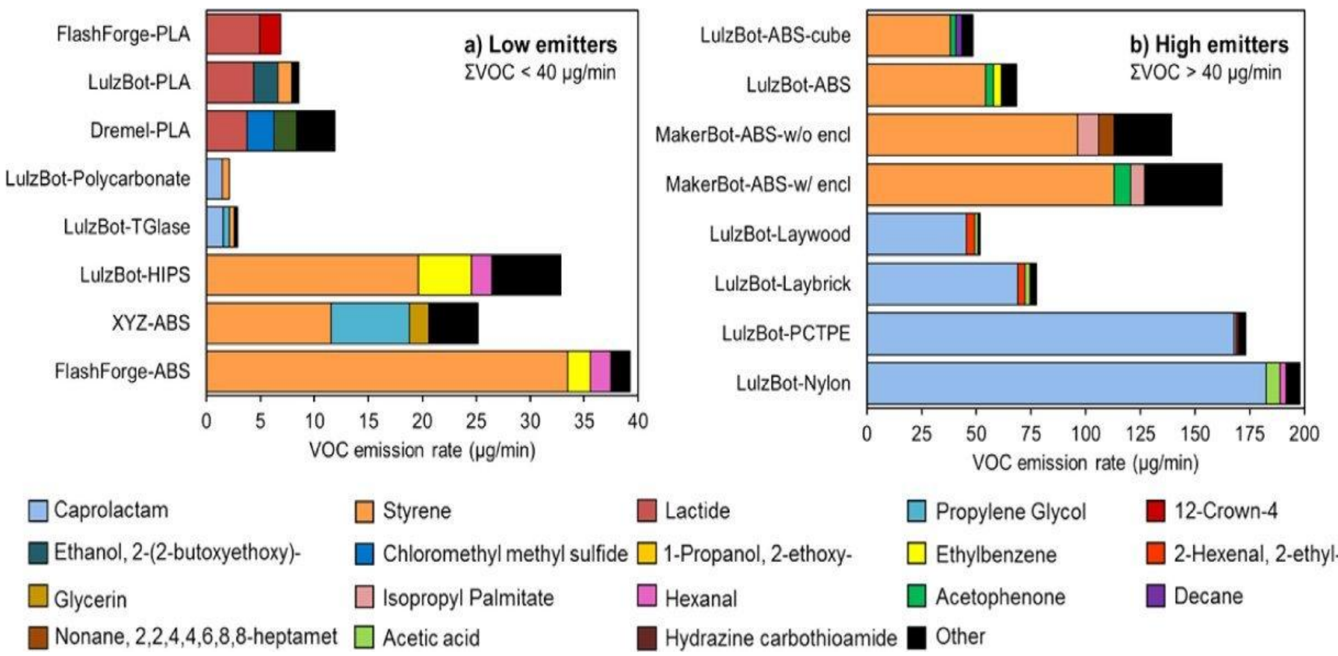


Figure 3: **Summary of estimates of individual speciated VOC and total VOC emission rates from each of the 16 printer and filament combinations.** [4]

Table 2: 3D Printing VOCs Overview

Compound	Sources	Health Impacts	Common Uses
Acetic Acid	Resin & FDM	Irritant	Vinegar, chemical reagent
Acetone	Resin & FDM	Impacts nervous system	Nail polish remover, PVC, solvent
Alpha-pinene	Resin & FDM	Generally safe	Fragrances, flavorings
Caprolactam	FDM	Sensitizer	Artificial textiles, nylon, resins
Cyclopentanone	Resin	Irritant	Fragrances, resins, adhesives
Dioxane	FDM	Reasonably anticipated to be a carcinogen	Inks, adhesives, solvent
Ethanol	Resin & FDM	Impacts nervous system	Solvent, fuel, liquor
Ethyl Acetate	FDM	Irritant	Nail polish remover, paints, adhesives
Ethyl Hexanol	Resin & FDM	Irritant	Plasticizer, fragrances, paints, resins

Ethyl Methacrylate	Resin	Sensitizer	Plexiglass, acrylic nails, dentures
Formaldehyde	Resin & FDM	Known to be a human carcinogen	Embalming, adhesives, resins
Hydroxypropyl Methacrylate	Resin & FDM	Sensitizer	Dental fillings, adhesives, resins
Isobornyl Acrylate	Resin & FDM	Sensitizer	Photoinitiator, inks, adhesives, resins
Isopropyl Alcohol	Resin & FDM	Respiratory irritant	Disinfectant, solvent
Lactide	PLA	Generally safe	PLA, drug delivery, packaging
Limonene	FDM	Generally safe	Fragrances, cleaners, solvent
Methyl Ethyl Ketone (MEK)	Resin & FDM	Impacts nervous system	Inks, paints, adhesives, resins, rubbers
Methyl Methacrylate	Resin & PLA	Sensitizer	Resins
Nonanaldehyde	Resin	Irritant	Fragrances, flavorings, polishes
Propylene Glycol	Resin & FDM	Generally safe	Antifreeze, food preservative, cosmetics
Styrene	Resin & ABS, HIPS	Reasonably anticipated to be a carcinogen	Styrofoam, latex, insulation, resins
Tetrachloroethylene	PLA, ABS, PC	Reasonably anticipated to be a carcinogen	Dry cleaning, degreaser, solvent
Toluene	Resin & FDM	Impacts nervous system	Gasoline additive, urethane, TNT
Xylene	Resin & FDM	Impacts nervous system	Concrete sealer, lubricant, solvent

Table 3: Minimum Risk Levels for Hazardous Substances

Formaldehyde			
Inhalation	Acute (1-14 days)	0.04 ppm	0.049 mg/m^3
Inhalation	Intermediate (4-364 days)	0.03 ppm	0.03684 mg/m^3
Inhalation	Chronic (365 days and longer)	0.008 ppm	0.00982 mg/m^3
Propylene Glycol			
Inhalation	Intermediate (4-364 days)	0.009 ppm	
Xylene, Mixed			
Inhalation	Acute (1-14 days)	2 ppm	
Inhalation	Intermediate (4-364 days)	0.6 ppm	

Inhalation	Chronic (365 days and longer)	0.05 ppm	

Table 4: PM Screening Values: World Health Organization Particulate Matter Air Quality Guidelines (AQG)

PM Air Pollutant Metric	WHO
PM10	45 µg/m3 (24-hour) † 15 µg/m3 (annual)
PM2.5	15 µg/m3 (24-hour) † 5 µg/m3 (annual)

† These screening levels reflects the numeric value of the WHO AQGs for 24-hours

Table 5: Most Notable Volatile Organic Compounds during Printing Process

VOCs	Typical filament / Process source	Health / Toxicity notes	Relative abundance / Behavior
Styrene	ABS, HIPS (high-temperature styrenic polymers)	Irritant; [illegible] (IARC Group [illegible]) [illegible] effects at high exposures.	Frequently among the highest VOCs [illegible] nozzle temperatures.
Caprolactam	Nylon (PA) filaments	Eye/respiratory irritant; [illegible]	Dominant VOC for many nylon prints (often the major contributor to TVOC from nylon).
Formaldehyde	Detected from ABS, PLA, nylon and some resins	Known human carcinogen; [illegible] chronic exposure [illegible]	Frequently detected across filament [illegible]
Acetaldehyde	ABS, PLA, nylon, some resins	Irritant; possible carcinogen (IARC Group 2B).	Commonly detected in many studies; often co-emitted with formaldehyde and other aldehydes.
Benzene	Trace in some ABS/HIPS/resins (esp. certain filaments/resins)	Known human carcinogen (IARC Group 1) [illegible]	Usually low concentrations [illegible] poorly ventilated spaces.
Toluene / Ethylbenzene / Xylenes (BTEX)	ABS, some resins, certain composite filaments	CNS effects, irritation; [illegible] carcinogenic [illegible]	Detected variably depending on filament [illegible] part of a complex VOC mix [illegible]
Lactide / lactic acid derivatives	PLA filaments (thermally-degraded PLA emissions)	Respiratory irritation [illegible]	Often reported [illegible] characteristic PLA [illegible] component of PLA VOC fingerprint).

Methyl methacrylate (MMA)	Some PLA variants, PMMA (clear resins, photopolymers) (post-processing)	Irritant to eyes/respiratory; occupational exposure limits	Observed in PLA and resin workflows; levels depend on chemistry and post-processing
Benzaldehyde / other oxygenated aromatics	Various filaments and resins	Irritants; some cause sensitization and mucosal effects	Frequently detected as part of the complex VOC profile; varies by filament packages and additive
Other aldehydes & ketones (e.g., acetone derivatives)	Multiple filaments/resins	Irritants; some have CNS effects at higher conc.	Common minor components; often co-... formaldehyde/acetaldehyde.

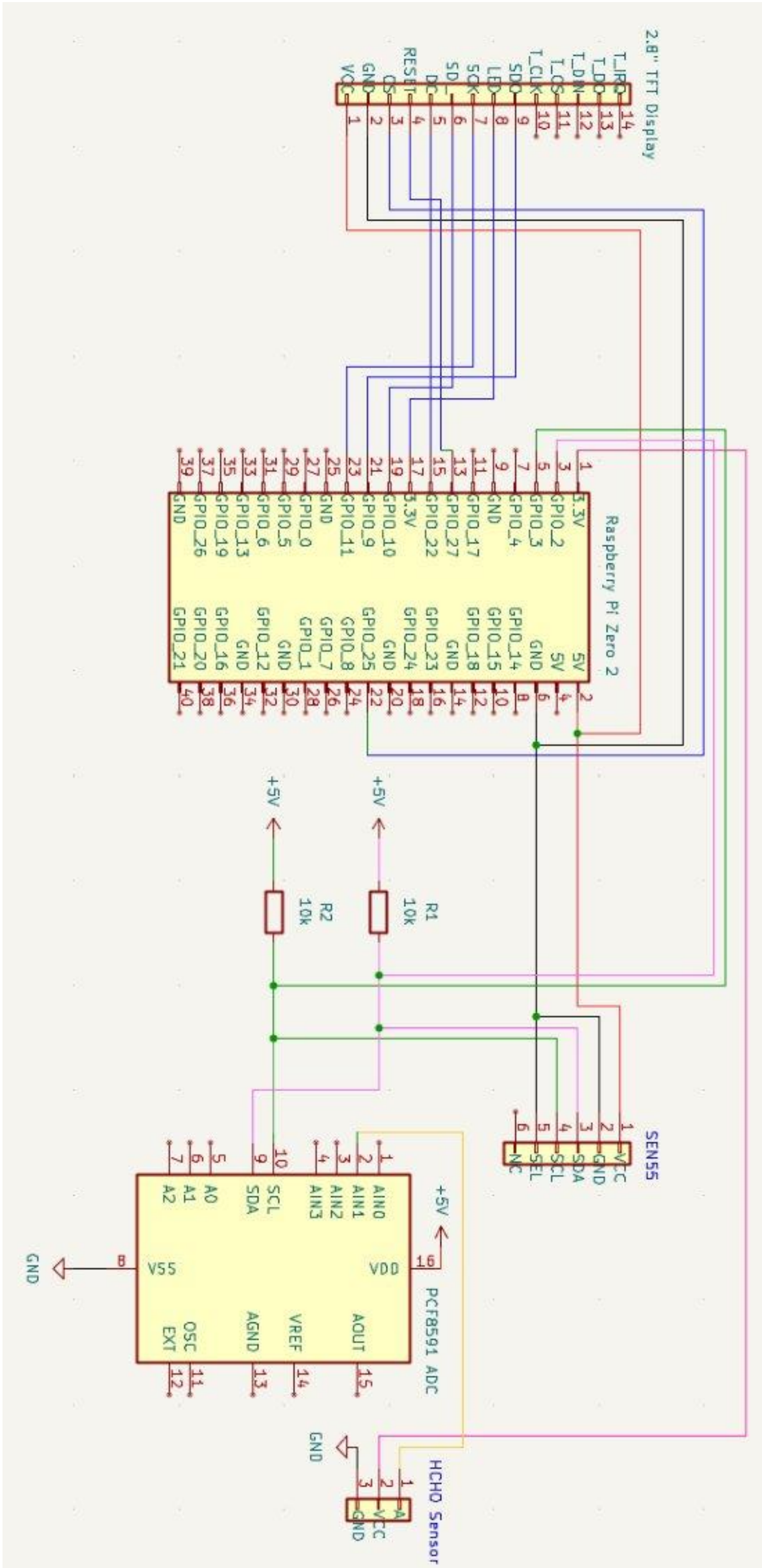
Project Management

Technical Information

Table 6: Design Objectives

Customer Requirements	Description	Priority
Data Logging and Analytics	Record data and analyze air quality	1
Following of Regulations	The concentration limit must adhere to current industry standard	1
Life Cycle	How long will this run with acceptable accuracy	4
Portability	Compact and can fit seamlessly into any 3D printing set up	2
Alert Mechanism	Trigger warnings when emission concentration reaches the limit	3
Cost	Affordable price for the general publics and education users, being able to compete with current product	2
Real-Time Monitoring	Continuous monitor of air quality throughout the printing process	1
Detection of Emission	Identify VOCs, ultrafine particles (PM1.0, PM2.5, PM10), Formaldehyde	1
Accuracy	Accurately detect the concentration so the alarm can go off and let user know	1

Note: For priority, 1 is the most important, and 4 is the least important.



2.8" TFT Display
Raspberry Pi Zero 2
SEN55
PCF8591 ADC
HCHO Sensor
R1
10k
R2
10k
+5V
GND

Figure 4: Prototype Schematics: Sensors & Data Processing Unit

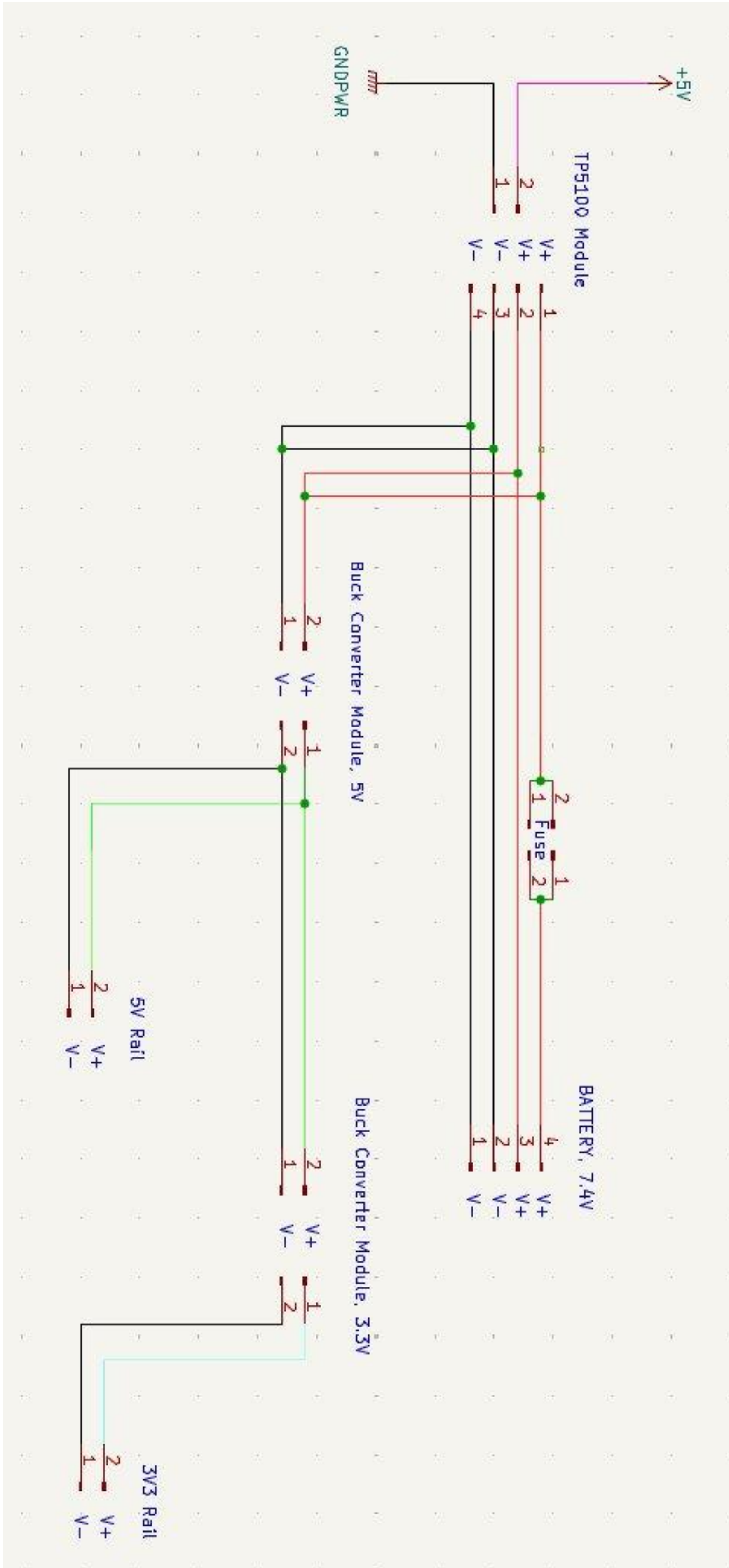

+5V
GNDPWR
TP5100 Module
BATTERY, 7.4V
Fuse
Buck Converter Module, 5V
5V Rail
Buck Converter Module, 3.3V
3V3 Rail

Figure 5: Prototype Schematics: Power Supply and Battery Management Unit

ASSUMPTIONS AND CALCULATIONS

- The Air monitor system mostly deals with airflow monitoring inside an electronic insulated enclosure. Therefore, computer fluid dynamics as well as thermodynamics principles will be adequate for the report of precise calculations of the following parameters influencing the Airflow of a system:
- **The Volume flow rate:** defined in CFM represents the volume of air per minute an external device can produce per minute in an environment. This parameter will be used to define the restrictions and the necessary RPM needed for the fan choice.
- **Dynamic Pressure:** it represents the kinetic energy per volume gained by the airflow. It will be useful to identify the pressure the airflow would gain before being brought to zero velocity without any losses.
- **Flow Velocity: It's** necessary to determine the air inlet flow velocity to meet the velocity of airflow requirements of the sensors SEN 55 which needs a maximum airflow velocity less than 1m/s.

ASSUMPTION:

- ISOCHORIC PROCESS PRESENT IN THE ENCLOSURE
- TRANSIENT STATE PROCESS DUE TO TIME DEPENDENT PARAMETERS IN TEMPERATURE AND PRESSURE
- HEAT SOURCES FROM RASPBERRY PI, SENSORS, AND BATTERY
- INTAKE PRESSURE AT ENVIRONMENT PRESSURE BUT MAY VARY

CONSTRAINTS AND PARAMETERS

- Total heat: Q = dissipated power from raspberry pi + Battery
- Recommended Airflow condition for SEN 55: Velocity < 1ms

CALCULATIONS

Necessary RPM = 4000 RPM

Average Dynamic Pressure:

Dp= (½)x ρ v^2 = 7.6 pa

Outlet velocity (Determined using Energy conservation equation)

$$v_{out}(t) = \frac{\dot{m}_{in}(t) + V\,\frac{\rho(t)}{T(t)}\,\frac{dT}{dt}}{\rho_{out}(t)\,A_{out}}.$$

From the simulation results the outlet velocity converges towards 1.4m/s

$$\boxed{\dot{m}_{out}(t) = \dot{m}_{in}(t) + V\,\frac{\rho(t)}{T(t)}\frac{dT}{dt}.}$$

Exposure Concentration:

C = ER (AV_m)(1N_m)

***C* (µg/m³) -**estimated exposure concentration;

ER (µg/h) -emission rate determined from the chamber study;

***V_m* (m^3) -**modeled volume;

***A:* -**number of 3D printers in the modeled volume

***N_m* (h^{-1}) –** the air change rate of the modeled volume.

Continuity equation:

$$\partial\rho/\partial t+\nabla\cdot(\rho v)=0$$

Q - the air flow (m^3)

V -the flow velocity, (m/s)

A- the cross-sectional area of the path taken. (m^2)

For a 6-hour resin printing session with 15 minutes of cleaning and 30 minutes of curing, the Formaldehyde emitted over the session is:

C= (6 hours × 0.006 mg/hr) + (0.25 hours × 0.012 mg/hr.) + (0.5 hours × 0.012 mg/hr.) = 0.045 mg.

The average US room size: 11 x 12 x 8 [ft] which is 1056 [ft^3] or 321.87 [m^3] so the concentration for Formaldehyde is 0.0015 mg/ m^3 with no air change rate per hour. For the success of sensor performance in sensing emissions, we may use different parameters related to fluid mechanics to maximize the flow of emission into the cavity of the gas detectors. Then, to find the necessary dimensions of the inlet pressure, we may use the continuity equation.

With the (Q been the air flow, V the flow velocity, A the cross-sectional area of the path taken. For a drop test, the identification of axial stresses present on the housing should be considered with the formula

Converting PPM to $\frac{mg}{m^3}$

Concentration in $\frac{mg}{m^3}$

$$\frac{(molecular\ weight) \times (concentration\ in\ ppm)}{24.45}$$

The range of concentration of the MEMS HCHO sensor in $\frac{mg}{m^3}$

$$\frac{(1.0078\times2+12.011+15.999)\times3}{24.45} = 3.684\ mg/m^3$$

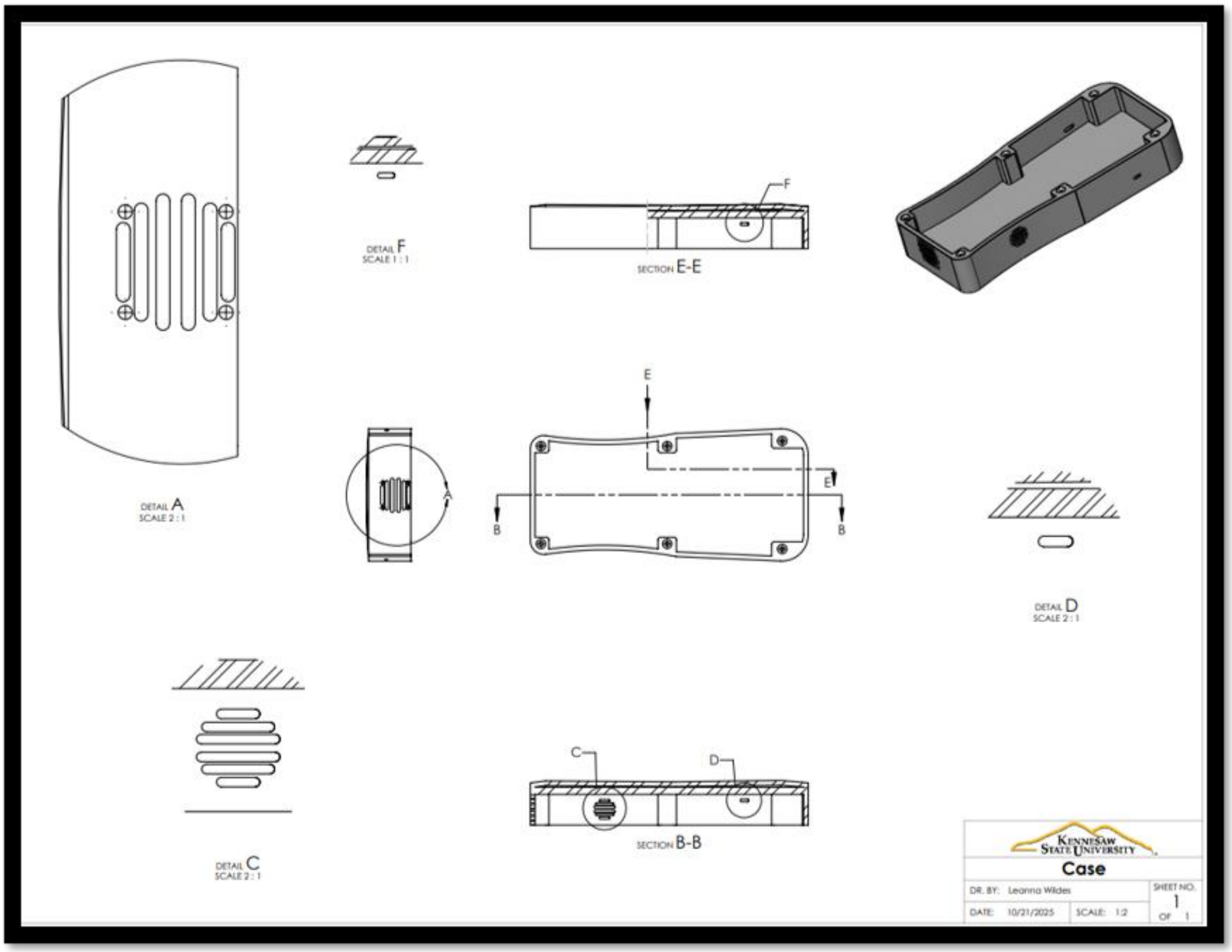


Figure 6: Part Drawing of Bottom of the Case

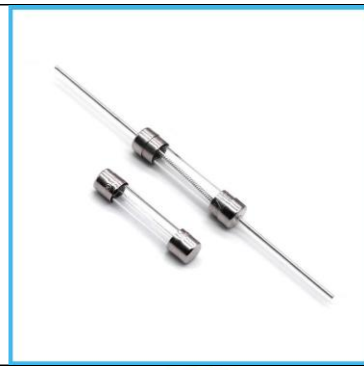
Power Supply & Battery Unit — Testing and Validation

Observation

The power supply subsystem consists of a **7.4 V / 10 000 mAh Li-ion pack**, a **TP5100 dual-cell charging module**, and **buck converters** for 5 V and 3.3 V regulation. The setup must safely charge the battery, deliver regulated voltages under dynamic load, and protect the system from overcurrent via a 3 A / 5 V fuse.

Test Equipment

Equipment	Appearance	Description
Digital Multimeter (DMM)		Used to measure voltage, current, and resistance throughout the power circuit. It verifies supply levels, detects voltage drops across loads, and confirms proper continuity in wiring.
12v DC Adapter (2A rated)		Provides a stable input source to power the TP5100 charging module. It emulates the external power adapter that would be used during system charging and testing.
Oscilloscope		Measures and visualizes voltage ripple, transient response, and noise on the DC outputs. It ensures that the buck converters maintain smooth and stable output suitable for sensitive electronics like sensors and microcontrollers.

3 A Fuse		Provides overcurrent protection and safety isolation during testing. The fuse prevents damage to components in case of wiring faults or short circuits
IR Thermometer		Monitors temperature rise on critical components such as the TP5100, buck converters, and battery pack during load testing. This ensures that the design stays within safe thermal limits.
Resistive Loads		Used to simulate the system's power consumption under different operating conditions. These loads help evaluate voltage regulation, thermal behavior, and runtime performance of the power supply.

Test Procedure

1. **Initial Verification:** Check wiring continuity, polarity, and module orientation.
2. **Charging Test:** Apply 12 V input to TP5100 and observe LED indicators; measure charge voltage at 8.4 V ± 0.05 V during CV stage.
3. **Battery Output Check:** Disconnect supply and record pack voltage between 6.5–8.4 V.
4. **Buck Converter Setup:** Calibrate 5 V converter output to 5.10 V and 3.3 V converter to 3.31 V under no load.
5. **Load Testing:** Incrementally increase current draw (0.5 A → 3 A) while monitoring voltage drop, ripple, and thermal rise.
6. **Protection Validation:** Simulate short circuit at output and verify 3 A fuse trip behavior and recovery.
7. **Charging While Operating:** Power the TP5100 and load simultaneously to confirm system-level stability during charge–discharge overlap.

System Test Outcomes

- TP5100 required **12 V input** for effective 2S charging; achieved full 8.4 V output.
- Battery maintained **7.6 V nominal** under 15 W system load.
- 5 V rail stayed between **5.06–5.12 V**, 3.3 V rail between **3.28–3.33 V**.
- Ripple measured < 40 mV (5 V rail) and < 25 mV (3.3 V rail).
- Power system supported **Raspberry Pi + sensors for 1.9 h runtime** on full charge.
- Thermal rise on buck converter < 24 °C above ambient.
- Fuse successfully isolated circuit on overload, confirming overcurrent protection.

Performance Benchmarks

- Input voltage: **12 V ± 5%**
- Charging voltage: **8.4 V ± 0.05 V**
- Regulated outputs: **5.10 ± 0.05 V**, **3.31 ± 0.03 V**
- Ripple: **≤ 40 mV p-p**
- System runtime: **≈ 1.9 h @ 15 W**
- Thermal limit: **< 45 °C**
- Overcurrent trip: **3 A fuse activates < 2 s**

Conclusions

The power circuit demonstrated **stable operation, safe charging, and adequate regulation** for all modules. Voltage rails remained within tolerance, with minimal ripple and acceptable thermal performance. The design meets electrical reliability and safety expectations for portable operation.

Formaldehyde Sensor Testing Procedure

The IAQ system will be turned on and let run for at least one hour to warm up. The interior volume, the ambient temperature, and relative humidity of the test chamber will be recorded. After one hour, data from the sensor can be recorded for the Zero calibration test. For the Span calibration, 6 formaldehyde concentration will be used for this test: 0.02, 0.05, 0.1, 0.5, 1, 3 ppm. The sensor will be exposed to the formaldehyde source over a 30-minute duration, follow by a 10-minute of flushing the chamber with fresh air. Repeat the process for all the concentrations starting with the lowest one. The same test will be repeated three times to insure the accuracy of the sensor and the repeatability of the test.

Results/Analysis

Formaldehyde Sensor Accuracy

Chamber Volume in m^3:

Chamber Temperature (degree C):

Chamber Relative Humidity:

Table X.

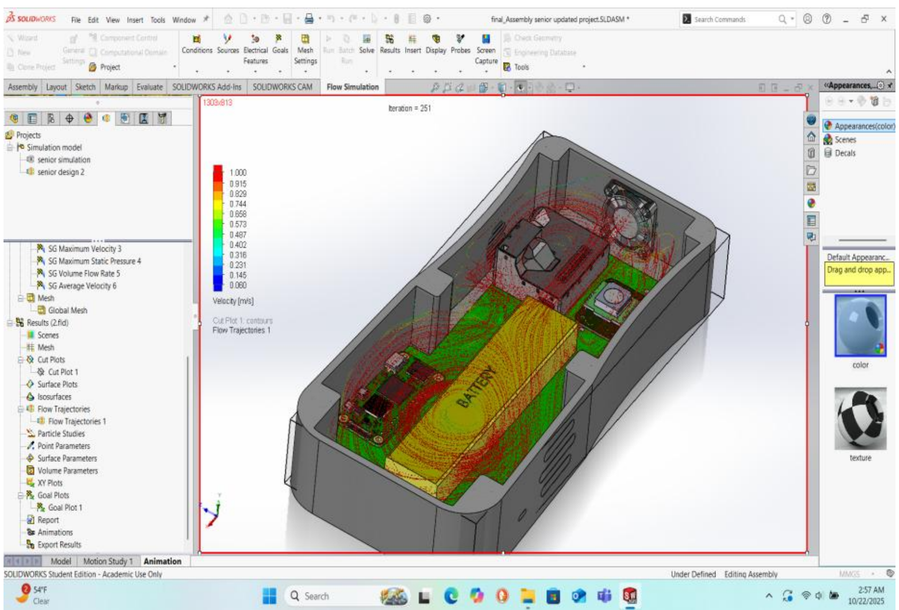
True Concentration		Measured Concentration (mg/m^3)		
		Test 1	Test 2	Test 3
ppm	mg/m^3			
0	0			
0.02	0.02456			
0.05	0.0614			
0.1	0.12			
0.5	0.61			
1	1.23			
3	3.68			

Particulate Matter Sensor Accuracy

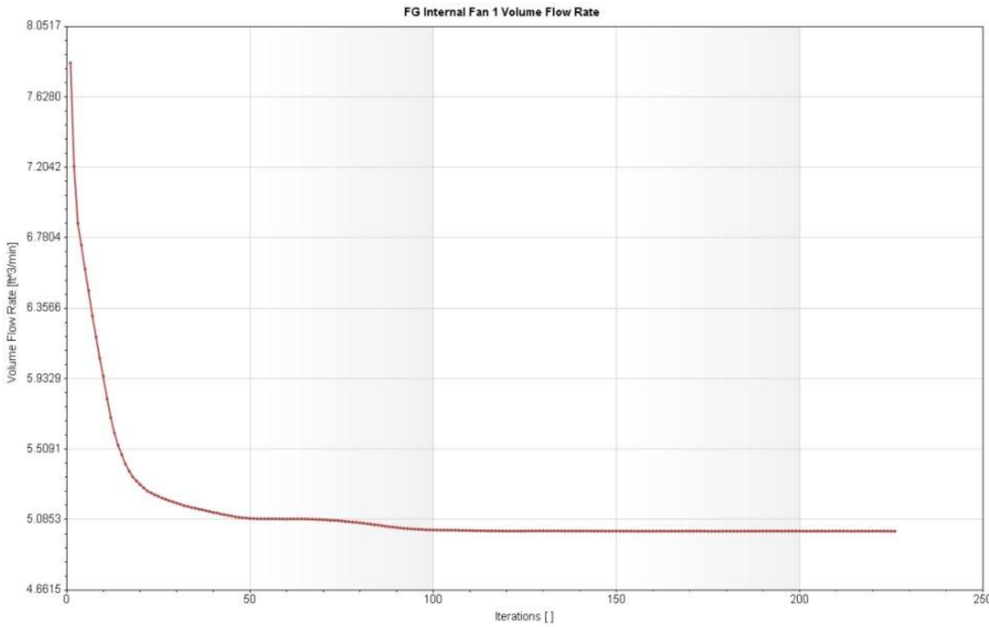
Computer Fluid Dynamic Simulation

The air monitor projects use the environment pressure to optimize the airflow in the enclosure using an axial fan with a velocity of 5000 RPM at maximum power. Due to the power dissipated by the battery, Raspberry pi Zero 2 and the sensors, the enclosure might face an increase of temperature over time equivalent to approximately 15W. Due to the change of pressure and temperature over time inside the enclosure, we consider the enclosure system to be in a transient state for the determination of the flow velocity and volume flow rate. The fan should produce an airflow to ensure the cooling process and the proper intake process of the environment flow into the system to favor the sensor chemical reaction to collect the different data and concentration of VOC'S present in the flow system.

Our goal over this flow analysis is to validate the choice of the fan used to optimize the airflow in the system to prevent overheating and favor the collection of accurate concentrations and data of VOC'S present in the enclosure for the air monitor system. Using SolidWorks Flow simulation, we might present the various results issued from several iterations of the software considering an external simulation with the environment pressure as our boundaries for both the inlet and outlet in the system to approach real conditions.

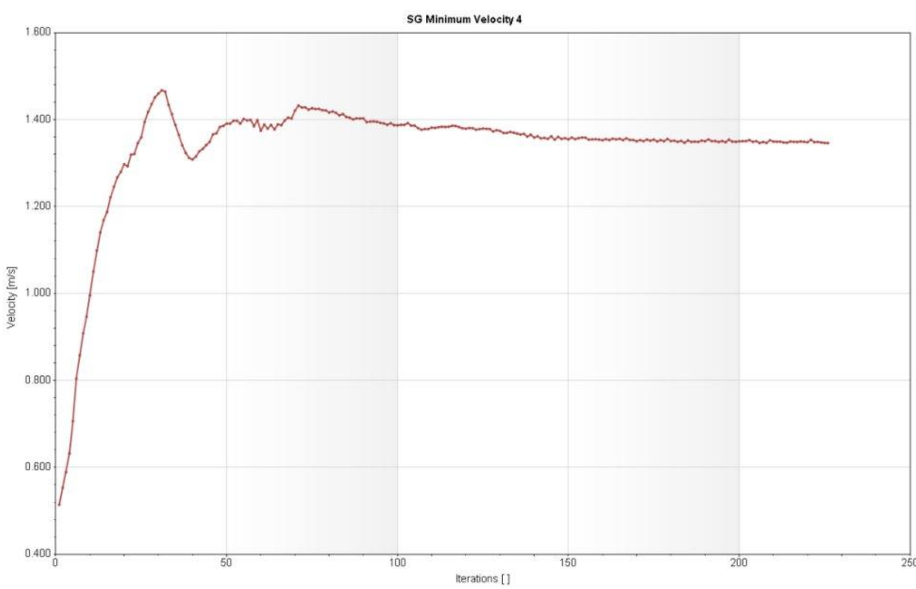


The plot indicates a flow velocity of 3m/s as the maximum value of flow produced by the fan at its optimum power along the axis of flow. Due to the static pressure that counters the airflow at the side, the velocity of the flow decreases between 0.915 and 1 m/s that fairly meet the sensors SEN55 requirement for the airflow.



GRAPH 1: Fan volume flow rate convergence

This plot shows the convergence of the maximum volume flow rate produced by the internal 30X30X8 axial fan while being subjected to the environment pressure in the inlet. As the number of iterations increases, the value of the flow rate becomes constant and converges to approximately 5 CFM at maximum performance.



GRAPH 2: Minimum flow velocity produced by the fan at maximum performance

This displays a flow velocity value issued from the Fan while subjected to a maximum static pressure (pressure preventing the airflow) in the enclosure. That value convergences to a velocity less than 1.4 m/s which is still ideal for the airflow in SEN55 and formaldehyde sensors

FUTURE FEA CONSTRAINTS

- **Tensile pull-out** of the screw boss (screw being pulled out of the plastic) — important if the fan or cable can pull on the screw or if the part gets dropped.
- **Local compression / crushing** under the screw head or washer — the plastic under the head can deform if the contact area is small or overtightened.
- **Shear** at the screw (if the fan applies lateral loads or the casing is bumped).
- **Fatigue / loosening from vibration** — the fan vibrates and cycles fast; repeated vibration can wear threads or back screws out.
- **Stress concentration / cracking** around mounting holes or at thin walls — sharp corners and thin walls concentrate stress.

Modelling and Assembly

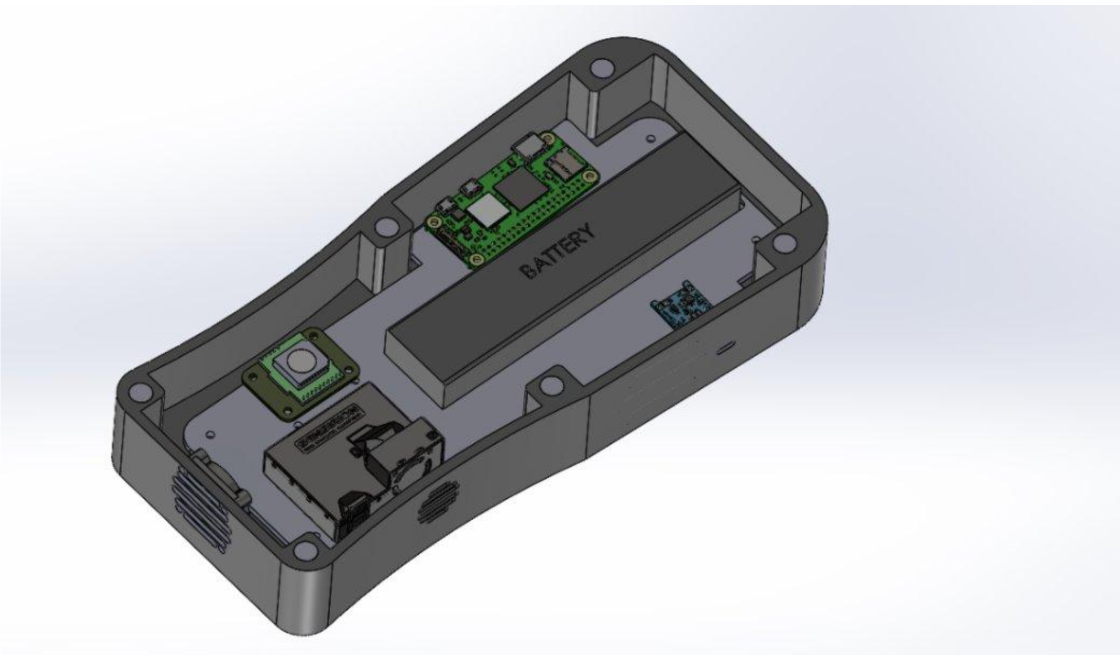


Figure 8: Detail assembly

The model describes our initial electronic and enclosure assembly to address the issue of heat exchange between the different components in the system and ease the interchangeability of the power source or battery, we propose a second system design that minimize heat inside the system and favor a greater configuration of the electronic system and enclosure assembly.

Figure 9: COMPLETE ASSEMBLY

Challenges

- Mold creation for mass production of housing.
- Improved electrical components for higher quality.
- Further research/testing on sensor capabilities and airflow improvements.
- Heat exchange and Battery interchangeability

Final Budget

Category	**Cost ($)**
3D printed case	
Sensirion SEN55	$27.38
Fermion MEMS HCHO sensor	$4.90
Raspberry Pi Zero 2W	$18.00
Adafruit 12 mm Square Tactile Buttons	$ 5.95
MakerHawk 3.7 V, 8000 mAh LIPo Battery	$19.09
Soberton Inc. General Purpose Speaker	$1.95
2.8” TFT Display Resistive Touch Screen module	$16.39
Miscellaneous/Electronic Components	$60.36

Total:	$154.02

Recommendations for Future

For future development of this product, it is recommended that:

- Creating a mold for larger plastic components for possible mass production capabilities
- Installation of HEPA filters between sensor components to help alleviate build up and to also help reduce possible false readings for more accurate measurements
- Replace Raspberry Pi Zero 2 with another module of ESP32 to increase operating time.

Conclusion

In conclusion, the air monitoring system for detecting VOCs and formaldehyde in SLA and FDM printer environments is progressing well. Data collection and transmission via Raspberry Pi and sensors function effectively. With portability achieved and the display operational, the remaining task focuses on finalizing the enclosure for complete deployment. Further finite element analysis and drop testing will require 3d printing to make up a finished prototype to validate the optimization of the airflow and the case toughness.

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Appendix

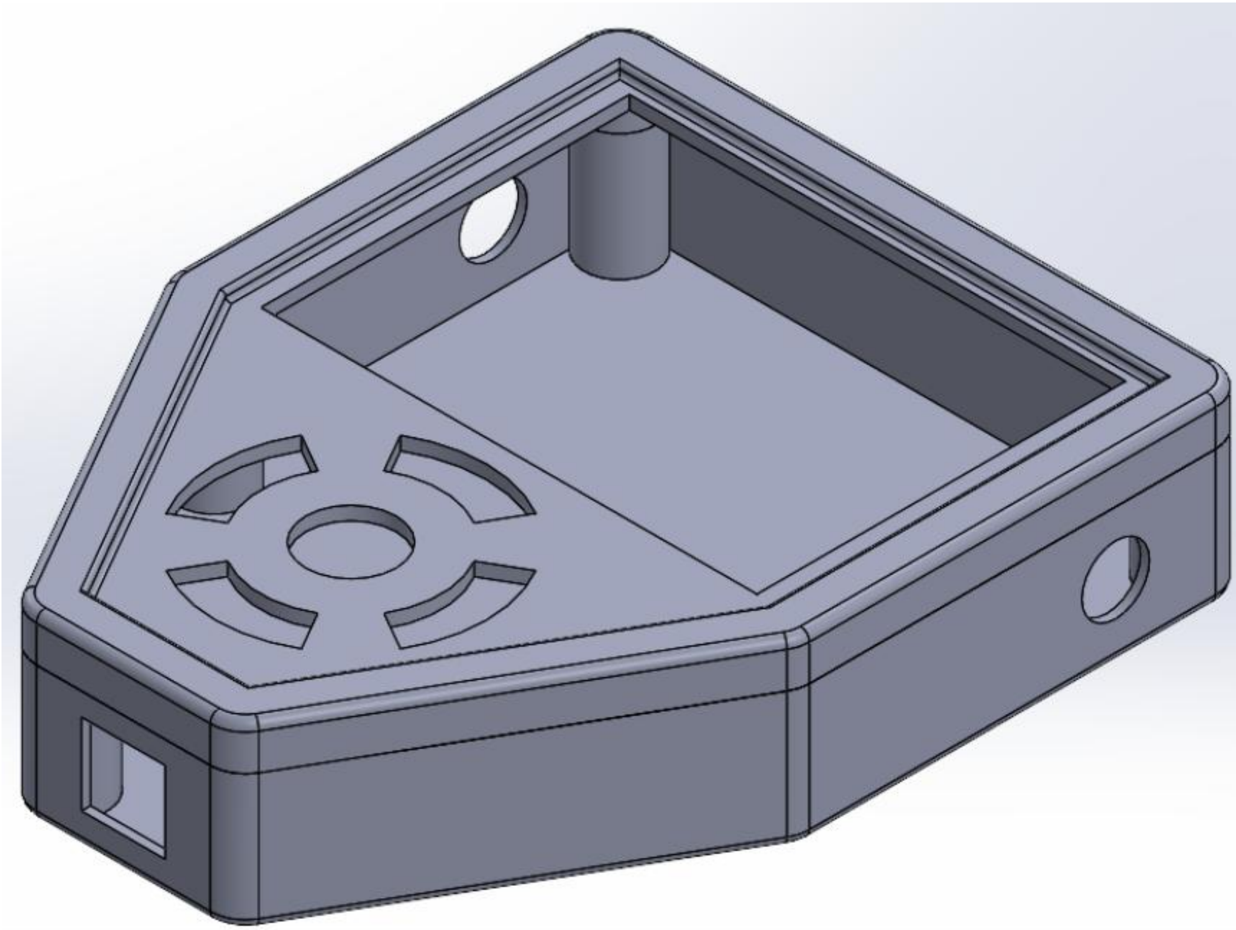

Image x: Previous design concept By: Forrest Austin

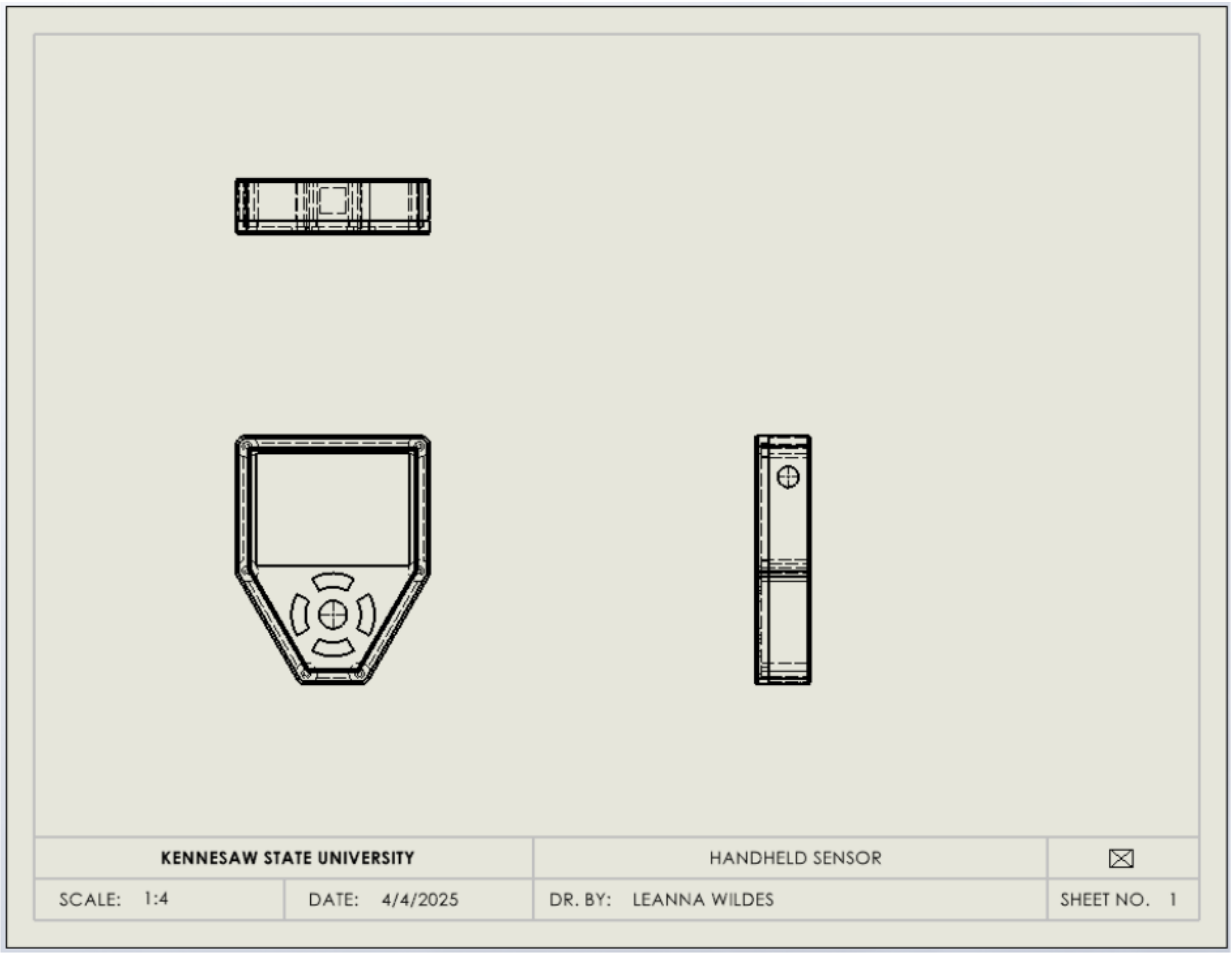

Image X: Previous design